Problem Set I – Solutions

Econometrics I - FGV EPGE

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The code for Problems 3, 5 and 6 is in R and is reproduced at the end of each problem. Simulations use `set.seed(2026)`, so the numbers below are exactly reproducible.

Problem 1

Throughout, K is a symmetric density, so $K(u) \geq 0$, $\int K(u)\,du = 1$ and $K(u) = K(-u)$, with $R(K) = \int K^2(u)\,du < \infty$. The workhorse is the change of variables $u = (z - x)/h$:

$$\mathbb{E}\left[\frac{1}{h}K\left(\frac{X_i - x}{h}\right)\right] = \frac{1}{h}\int K\left(\frac{z-x}{h}\right) f(z)\,dz = \int K(u) f(x + hu)\,du. \tag{1}$$

a) Each term $K\big((X_i - x)/h\big)$ is nonnegative and $1/(nh) > 0$, so $\hat{f}_n(x) \geq 0$. For the integral, use $u = (X_i - x)/h$ in each term, so $dx = -h\,du$ and the limits swap:

$$\int_{-\infty}^{\infty} K\left(\frac{X_i - x}{h}\right) dx = h\int_{-\infty}^{\infty} K(u)\,du = h \quad \Longrightarrow \quad \int_{-\infty}^{\infty} \hat{f}_n(x)\,dx = \frac{1}{nh}\sum_{i=1}^{n} h = 1.$$

So $\hat{f}_n$ is itself a density, for every sample and every n.

b) Since the X_i are identically distributed, $\mathbb{E}[\hat{f}_n(x)]$ equals (1). Using $\int K = 1$,

$$\mathbb{E}[\hat{f}_n(x)] - f(x) = \int K(u)\big[f(x + hu) - f(x)\big]\,du.$$

For each fixed u, $f(x + hu) \to f(x)$ as $h \to 0$, because f is continuous at x. Let $M = \sup_z f(z) < \infty$. The integrand is bounded by $2M\,K(u)$, which is integrable. By dominated convergence (along any sequence $h \to 0$), the integral goes to zero:

$$\mathbb{E}[\hat{f}_n(x)] = f(x) + o(1).$$

Boundedness of f is what lets us use kernels with unbounded support (such as the Gaussian): far-away points get little weight, and they cannot carry arbitrarily large density.

c) By independence,

$$\mathrm{Var}\big(\hat{f}_n(x)\big) = \frac{1}{n}\mathrm{Var}\left(\frac{1}{h}K\left(\frac{X_i - x}{h}\right)\right) = \frac{1}{n}\,\mathbb{E}\left[\frac{1}{h^2}K^2\left(\frac{X_i - x}{h}\right)\right] - \frac{1}{n}\Big(\mathbb{E}[\hat{f}_n(x)]\Big)^2.$$

For the first term, the same change of variables gives

$$\mathbb{E}\left[\frac{1}{h^2}K^2\left(\frac{X_i - x}{h}\right)\right] = \frac{1}{h}\int K^2(u) f(x + hu)\,du = \frac{1}{h}\Big(f(x)R(K) + o(1)\Big),$$

where the last step is dominated convergence again, now with dominating function $MK^2(u)$, integrable because $R(K) < \infty$. For the second term, item b) gives $\big(\mathbb{E}[\hat{f}_n(x)]\big)^2 = (f(x) + o(1))^2 = O(1)$. Putting the two together,

$$\mathrm{Var}\big(\hat{f}_n(x)\big) = \frac{f(x)R(K)}{nh} + o\left(\frac{1}{nh}\right) + O\left(\frac{1}{n}\right) = \frac{f(x)R(K)}{nh} + o\left(\frac{1}{nh}\right),$$

because $\frac{1/n}{1/(nh)} = h \to 0$, so $O(1/n) = o(1/(nh))$.

d) The mean squared error splits into variance plus squared bias:

$$\mathbb{E}\Big[\big(\hat{f}_n(x) - f(x)\big)^2\Big] = \mathrm{Var}\big(\hat{f}_n(x)\big) + \Big(\mathbb{E}[\hat{f}_n(x)] - f(x)\Big)^2 = \underbrace{\frac{f(x)R(K)}{nh} + o\left(\frac{1}{nh}\right)}_{\to 0 \text{ since } nh \to \infty} + \underbrace{o(1)}_{\text{since } h \to 0} \longrightarrow 0.$$

By Markov's (Chebyshev's) inequality, for any $\varepsilon > 0$,

$$\mathbb{P}\big(|\hat{f}_n(x) - f(x)| > \varepsilon\big) \le \frac{\mathbb{E}\big[(\hat{f}_n(x) - f(x))^2\big]}{\varepsilon^2} \longrightarrow 0,$$

so $\hat{f}_n(x) \xrightarrow{p} f(x)$. Each condition does one job: $h \to 0$ kills the bias and $nh \to \infty$ kills the variance.

e) Let

$$r(t) \equiv f(x+t) - f(x) - f'(x)\,t - \tfrac{1}{2}f''(x)\,t^2.$$

By (1), symmetry ($\int uK(u)\,du = 0$) and unit variance ($\int u^2K(u)\,du = 1$),

$$\mathbb{E}[\hat{f}_n(x)] = \int K(u)\Big[f(x) + f'(x)hu + \tfrac{1}{2}f''(x)h^2u^2 + r(hu)\Big]du = f(x) + \frac{h^2}{2}f''(x) + \int K(u)\,r(hu)\,du.$$

It remains to show that $\int K(u)r(hu)\,du = o(h^2)$. Fix $\varepsilon > 0$. Because f'' is continuous at x, there is a $\delta > 0$ such that f is twice differentiable on $(x-\delta, x+\delta)$ and $|f''(x+t) - f''(x)| \le \varepsilon$ whenever $|t| < \delta$. Split the integral at $|u| = \delta/h$.

Near x ($|hu| < \delta$). Taylor's theorem with the Lagrange remainder gives $f(x+t) = f(x) + f'(x)t + \frac{1}{2}f''(\xi)t^2$ for some ξ between x and $x+t$, so $|r(t)| = \frac{1}{2}|f''(\xi) - f''(x)|\,t^2 \le \frac{\varepsilon}{2}t^2$. Hence

$$\left|\int_{|u|<\delta/h} K(u)\,r(hu)\,du\right| \le \frac{\varepsilon}{2}h^2\int u^2K(u)\,du = \frac{\varepsilon}{2}h^2.$$

Far from x ($|u| \ge \delta/h$). Here we only use $f \le M$: $|r(hu)| \le 2M + |f'(x)|h|u| + \frac{1}{2}|f''(x)|h^2u^2$. On this region $1 \le (hu/\delta)^2$ and $h|u| \le h^2u^2/\delta$, so

$$\left|\int_{|u|\ge\delta/h} K(u)\,r(hu)\,du\right| \le h^2\left(\frac{2M}{\delta^2} + \frac{|f'(x)|}{\delta} + \frac{|f''(x)|}{2}\right)\int_{|u|\ge\delta/h} u^2K(u)\,du = o(h^2),$$

since the tail of the convergent integral $\int u^2K(u)\,du$ goes to zero as $\delta/h \to \infty$.

Together, $\limsup_{h\to 0} h^{-2}\left|\int K(u)r(hu)\,du\right| \le \varepsilon/2$ for every $\varepsilon > 0$, so the remainder is $o(h^2)$ and

$$\mathbb{E}[\hat{f}_n(x)] = f(x) + \frac{h^2}{2}f''(x) + o(h^2).$$

f) The leading bias is $\frac{h^2}{2}f''(x)$, so its sign is the sign of the curvature of f at x. The estimator replaces $f(x)$ by a weighted average of f over a window of width about h around x.

- Where f is locally convex ($f''(x) > 0$, e.g. in the tails of a bell-shaped density or at a trough), the average of f around x is above $f(x)$: the bias is positive.
- Where f is locally concave ($f''(x) < 0$, e.g. at a mode), the average is below $f(x)$: the bias is negative. Kernel estimates flatten peaks and fill in valleys.
- At an inflection point ($f''(x) = 0$) the leading bias vanishes, and the bias is of smaller order than h^2.

The slope $f'(x)$ plays no role, because a symmetric kernel puts equal weight on both sides of x, so a linear trend averages out. The bias grows with both $|f''(x)|$ and h^2. Problem 6 shows this bias at work.

Problem 2

a) Stack the data as

$$V = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & X_1 - x_0 \\ \vdots & \vdots \\ 1 & X_n - x_0 \end{pmatrix}, \qquad W = \operatorname{diag}\left(\frac{1}{K_h(X_1 - x_0)}, \ldots, \frac{1}{K_h(X_n - x_0)} \right),$$

and let $\theta = (\alpha, \beta)'$. Then $W^{-1} = \operatorname{diag}\big(K_h(X_i - x_0)\big)$ and the objective is

$$S(\theta) = \sum_{i=1}^{n} K_h(X_i - x_0)\big[Y_i - \alpha - \beta(X_i - x_0)\big]^2 = (V - Z\theta)'W^{-1}(V - Z\theta).$$

This is a quadratic function of θ with positive semidefinite Hessian $2Z'W^{-1}Z$, so the first-order condition is sufficient:

$$\frac{\partial S}{\partial \theta} = -2Z'W^{-1}(V - Z\theta) = 0 \quad \Longrightarrow \quad \begin{pmatrix} m_{LL}(x_0) \\ m'_{LL}(x_0) \end{pmatrix} = (Z'W^{-1}Z)^{-1}Z'W^{-1}V,$$

provided $Z'W^{-1}Z$ is invertible, which needs at least two distinct values of X_i with positive weight. This is the GLS formula with "error covariance matrix" W: an observation far from x_0 is treated as if it were very noisy (variance $1/K_h(X_i - x_0)$ is large), so it gets little weight.

Two remarks. First, W exists only if every weight is strictly positive, as with the Gaussian kernel. With a compactly supported kernel, the observations with zero weight do not enter $S(\theta)$ at all, so we drop

them and define V, Z, W on the remaining ones. Second, writing $S_j = \sum_i K_h(X_i - x_0)(X_i - x_0)^j$ and $T_j = \sum_i K_h(X_i - x_0)(X_i - x_0)^j Y_i$, the 2×2 inverse gives the closed form

$$m_{LL}(x_0) = \frac{S_2 T_0 - S_1 T_1}{S_0 S_2 - S_1^2}, \qquad m'_{LL}(x_0) = \frac{S_0 T_1 - S_1 T_0}{S_0 S_2 - S_1^2}.$$

b) Multiplying all weights by the same constant does not change the estimator (the constant cancels in $(Z'W^{-1}Z)^{-1}Z'W^{-1}V$). So we can drop the $1/h$ in K_h and use weights $K\big((X_i - x_0)/h\big)$. For fixed n and $h \to \infty$, each $(X_i - x_0)/h \to 0$, so each weight tends to $K(0) > 0$ (for a kernel continuous at zero). Then $W^{-1} \propto I_n$ in the limit, and, since $Z'Z$ is invertible,

$$\begin{pmatrix} m_{LL}(x_0) \\ m'_{LL}(x_0) \end{pmatrix} \longrightarrow (Z'Z)^{-1}Z'V.$$

This is OLS of Y_i on a constant and $X_i - x_0$. If $\hat{a} + \hat{b}X$ is the global OLS line, then $m_{LL}(x_0) \to \hat{a} + \hat{b}x_0$ and $m'_{LL}(x_0) \to \hat{b}$. With an infinite bandwidth every observation counts equally, so the local-linear estimator becomes the global linear regression, evaluated at x_0. (The local-constant estimator becomes the global constant, $\bar{Y}$.)

c) For the local-constant estimator, the lecture shows

$$\hat{m}(x) - m(x) \approx N\left(h^2\kappa_2 B(x),\ \frac{\sigma^2(x)R(K)}{nh\,f(x)}\right), \qquad B(x) = m'(x)\frac{f'(x)}{f(x)} + \frac{1}{2}m''(x).$$

- $h \to 0$ makes the bias vanish. The estimator averages Y_i over a window of width about h around x. If the window does not shrink, it keeps averaging $m(X_i)$ at points where $m(X_i) \neq m(x)$, and the estimator converges to a smoothed version of m, not to $m(x)$.
- $nh \to \infty$ makes the variance vanish. The number of observations in the window is about $n \cdot 2h \cdot f(x)$. If nh stays bounded, we are averaging a bounded number of noisy Y_i's, and the noise u_i never averages out, no matter how large n is.

Together they say: the window shrinks, but more slowly than $1/n$, so it still contains more and more observations.

d) Yes. The local-linear estimator is also a local average, now of a local line, so it faces the same tension: $h \to 0$ for its bias to vanish and $nh \to \infty$ for its variance to vanish. Under these conditions (and $nh^5 = O(1)$),

$$\sqrt{nh}\left(m_{LL}(x) - m(x) - h^2\kappa_2\frac{m''(x)}{2}\right) \xrightarrow{d} N\left(0,\ \frac{\sigma^2(x)R(K)}{f(x)}\right).$$

The main differences:

- **Same variance, different bias.** Both have the same asymptotic variance. The local-linear bias, $h^2\kappa_2 m''(x)/2$, drops the "design" term $h^2\kappa_2 m'(x)f'(x)/f(x)$ of the local-constant estimator. That

term appears when the regressors are unevenly spread around x ($f'(x) \neq 0$) and m has a slope: the local-constant fit then leans toward the side with more data. A local line absorbs the slope.

- **Exact for linear m.** If m is linear, $V = Z\theta + u$ exactly, so $\mathbb{E}[(m_{LL}, m'_{LL})' \mid X] = \theta$: no bias at all, for any h. The local-constant estimator is biased even then.
- **Boundary.** Near the edge of the support, the local-constant estimator has bias of order h (all neighbours lie on one side), while the local-linear estimator keeps bias of order h^2.
- **Derivative.** The local-linear estimator also delivers an estimate of $m'(x)$.
- **Cost.** It needs $Z'W^{-1}Z$ to be well conditioned, so it can be erratic where data are sparse (few distinct X_i in the window).

The Appendix proves that $h \to 0$ and $nh \to \infty$ are enough for $m_{LL}(x) \xrightarrow{p} m(x)$.

Problem 3

a) Karlan and Zinman ran a field experiment with a South African consumer lender: past borrowers received direct-mail offers with randomly assigned interest rates. The question is whether demand for small, high-interest loans to poor borrowers is price inelastic, as many microfinance practitioners assume when arguing that lenders can raise rates to cover costs without losing clients. The striking finding is that demand does slope down: fewer clients borrow when the rate is higher, and the response is steeper for rate increases above the lender's standard rates than for decreases. Loan size also responds much more to loan maturity than to the interest rate, especially for poorer clients, which points to liquidity constraints.

b) With $Y = \texttt{applied} \in \{0, 1\}$ and $X = \texttt{offer4}$,

$$p(x) \equiv \mathbb{P}(Y = 1 \mid X = x) = 1 \cdot \mathbb{P}(Y = 1 \mid X = x) + 0 \cdot \mathbb{P}(Y = 0 \mid X = x) = \mathbb{E}[Y \mid X = x],$$

so estimating the probability is a nonparametric regression problem. We use the Nadaraya–Watson estimator

$$\hat{p}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)},$$

a weighted share of applicants among clients offered rates near x. It always lies in $[0, 1]$.

c, d) The sample has $n = 53{,}810$ clients, $\hat{\sigma} = 2.472$, so Silverman's rule gives $h_S = 1.06 \cdot 2.472 \cdot 53810^{-1/5} = 0.297$. We use $h_S/5 = 0.06$ as the small bandwidth and $5h_S = 1.48$ as the large one. As in lecture 1, both kernels have unit variance:

$$K_{\text{Gauss}}(u) = \phi(u), \qquad K_{\text{Epa}}(u) = \frac{3}{4\sqrt{5}}\left(1 - \frac{u^2}{5}\right)\mathbf{1}\{|u| < \sqrt{5}\}.$$

Then a given h means the same amount of smoothing for both kernels, which is also what Silverman's rule assumes. (With the textbook Epanechnikov kernel on $[-1, 1]$, whose variance is $1/5$, the same h smooths about $\sqrt{5}$ times less.) Figure 1 shows the estimates. The grey dots are the raw application rates at each offered rate, with size proportional to the number of clients.

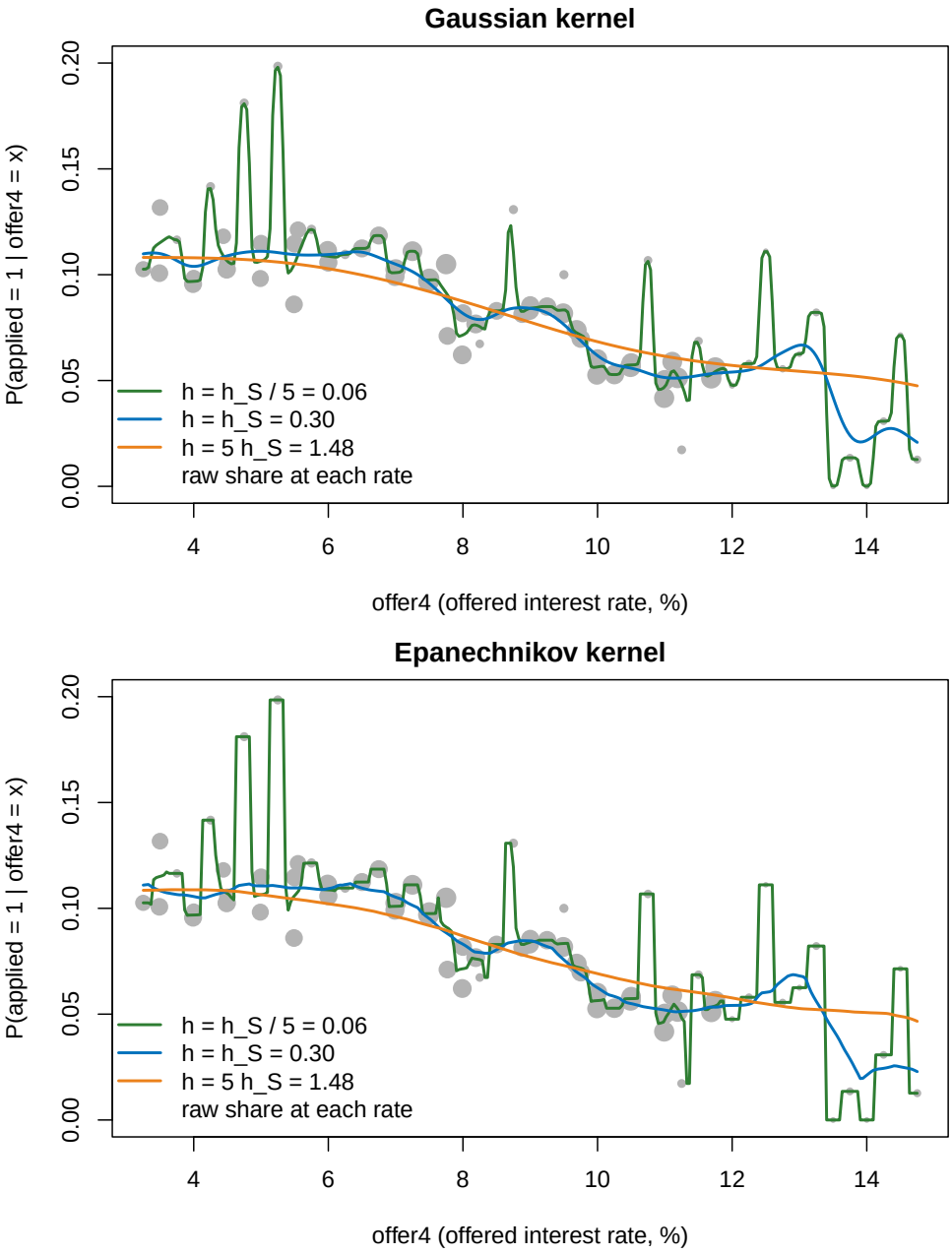


Figure 1: Nadaraya–Watson estimates of $\mathbb{P}(\texttt{applied} = 1 \mid \texttt{offer4} = x)$.

With Silverman's bandwidth, the application probability falls from about 10.4% at a rate of 4 to 8.2% at 8 and 5.4% at 12: demand slopes down, as in the paper.

e)

- **The bandwidth matters far more than the kernel.** With h_S the two curves are almost identical (they never differ by more than 0.0055). With $5h_S$ both are smooth, nearly linear declines.
- **Small bandwidth.** With $h_S/5$ both curves undersmooth: they chase the noise in thinly populated rates (the spikes near 5.25 and above 12, where few clients were offered each rate). Here the kernels look different. `offer4` takes only 70 distinct values, mostly on a grid of 0.25. The Epanechnikov kernel gives zero weight beyond $\sqrt{5} \cdot 0.06 \approx 0.13$, so at most points it averages only the clients at the nearest rate, and the curve is a sequence of flat steps that jump between rates. The Gaussian kernel never gives zero weight, so its curve moves smoothly from one rate to the next. The differences reach 0.077.
- **Large bandwidth.** With $5h_S$ the curves are smooth but biased where the true curve bends, and they are pulled toward the interior at the edges of the data (above 14, the large-h curve stays at about 5% while the data are lower), the boundary bias of the local-constant estimator from Problem 2.

Listing 1: Problem 3 (`code/p3_karlan.R`)

```
# Problem 3: Nadaraya-Watson estimate of P(applied = 1 | offer4 = x)
# Run from solutions/PS1/:  Rscript code/p3_karlan.R

df <- read.csv("../../problem_sets/PS1/karlan.csv")
Y <- df$applied
X <- df$offer4
n <- length(X)

# Bandwidths: Silverman's rule, a much smaller one and a much larger one
hS <- 1.06 * sd(X) * n^(-1/5)
H  <- c(small = hS / 5, silverman = hS, large = 5 * hS)
cat(sprintf("n = %d, sd(offer4) = %.3f, h_S = %.3f\n", n, sd(X), hS))

# Both kernels have unit variance (as in lecture 1), so a given h means the same
# amount of smoothing for both
k_gauss <- function(u) dnorm(u)
k_epan  <- function(u) ifelse(abs(u) < sqrt(5), 3 / (4 * sqrt(5)) * (1 - u^2 / 5), 0)

nw <- function(grid, X, Y, h, K) {
  sapply(grid, function(x) {
    w <- K((X - x) / h)
    if (sum(w) == 0) return(NA_real_)
    sum(w * Y) / sum(w)
  })
}

grid <- seq(min(X), max(X), length.out = 300)

# Raw frequencies at each distinct offered rate, to put on the plots
```

```
cells <- aggregate(applied ~ offer4, data = df, FUN = mean)
cells$n <- as.vector(table(df$offer4))

cols <- c("#2E7D32", "#0072BC", "#EB811B")
plot_kernel <- function(K, main, file) {
  est <- sapply(H, function(h) nw(grid, X, Y, h, K))
  pdf(file, width = 7, height = 4.5)
  par(mar = c(4.2, 4.2, 2, 1))
  plot(cells$offer4, cells$applied, pch = 16, col = "grey70",
       cex = 0.4 + 1.6 * sqrt(cells$n / max(cells$n)),
       ylim = c(0, 0.2), xlab = "offer4 (offered interest rate, %)",
       ylab = "P(applied = 1 | offer4 = x)", main = main)
  matlines(grid, est, lty = 1, lwd = 2, col = cols)
  legend("bottomleft", bty = "n", lwd = 2, col = c(cols, NA), pch = c(NA, NA, NA, 16),
         legend = c(sprintf("h = h_S / 5 = %.2f", H[1]), sprintf("h = h_S = %.2f", H[2]),
                    sprintf("h = 5 h_S = %.2f", H[3]), "raw share at each rate"))
  dev.off()
  est
}

est_g <- plot_kernel(k_gauss, "Gaussian kernel", "figures/p3_gaussian.pdf")
est_e <- plot_kernel(k_epan, "Epanechnikov kernel", "figures/p3_epanechnikov.pdf")

# A few numbers quoted in the solutions
at <- c(4, 8, 12)
idx <- sapply(at, function(a) which.min(abs(grid - a)))
tab <- rbind(gauss = round(est_g[idx, 2], 3), epan = round(est_e[idx, 2], 3))
colnames(tab) <- paste0("x=", at)
print(tab)
cat(sprintf("max | gauss - epan | at h_S: %.4f\n", max(abs(est_g[, 2] - est_e[, 2]), na.rm = TRUE)))
cat(sprintf("max | gauss - epan | at h_S/5: %.4f\n", max(abs(est_g[, 1] - est_e[, 1]), na.rm = TRUE)))
cat(sprintf("NA count Epanechnikov small h: %d\n", sum(is.na(est_e[, 1]))))
```

Problem 4

Let $X_1, \dots, X_n$ be i.i.d. $U[0, \theta]$ and $R_n = n\big(X_{(n)} - \theta\big)$.

a) Every $X_i \le \theta$ with probability one, so $X_{(n)} = \max_i X_i \le \theta$ and $R_n \le 0$.

b) The maximum is below t if and only if every observation is below t, so for $t \in [0, \theta]$, $\mathbb{P}(X_{(n)} \le t) =$

$\prod_i \mathbb{P}(X_i \le t) = (t/\theta)^n$. Hence

$$J_n(x,P) = \mathbb{P}\left(X_{(n)} \le \theta + \frac{x}{n}\right) = \begin{cases} 0, & x < -n\theta, \\ \left(1 + \dfrac{x/\theta}{n}\right)^n, & -n\theta \le x \le 0, \\ 1, & x > 0. \end{cases}$$

Fix $x \le 0$. For n large, $-n\theta \le x$, and the hint with $r = x/\theta$ gives $J_n(x,P) \to e^{x/\theta}$. For $x > 0$, $J_n(x,P) = 1$ for all n. So $J_n(x,P) \to J(x,P)$ at every x, with

$$J(x,P) = \begin{cases} e^{x/\theta}, & x \le 0 \\ 1, & x > 0 \end{cases} = \mathbb{P}(-\theta X \le x), \qquad X \sim \mathrm{Exp}(1),$$

since for $x \le 0$, $\mathbb{P}(-\theta X \le x) = \mathbb{P}(X \ge -x/\theta) = e^{x/\theta}$. That is, $R_n \xrightarrow{d} -\theta X$.

c) Write $\theta_n \equiv \theta(P_n)$ for the largest support point of P_n. Since $\varepsilon > 0$,

$$\mathbb{P}\Big(n\big(X_{(n),n} - \theta_n\big) \le -\varepsilon\Big) = \mathbb{P}\Big(X_{(n),n} \le \theta_n - \frac{\varepsilon}{n}\Big) \le \mathbb{P}\big(X_{(n),n} < \theta_n\big).$$

The sample maximum is below θ_n if and only if no draw hits the top point θ_n. Each draw hits it with probability $1/n$ (the n points are distinct and equally likely), independently, so

$$\mathbb{P}\big(X_{(n),n} < \theta_n\big) = \left(1 - \frac{1}{n}\right)^n \longrightarrow e^{-1}$$

by the hint with $r = -1$. Hence, for every $\varepsilon > 0$,

$$\limsup_{n\to\infty} \mathbb{P}\Big(n\big(X_{(n),n} - \theta(P_n)\big) \le -\varepsilon\Big) \le e^{-1}.$$

d) By definition, $J_n(-\varepsilon, P_n) = \mathbb{P}\big(n(X_{(n),n} - \theta(P_n)) \le -\varepsilon\big)$. Take $0 < \varepsilon < \theta$. Then $-\varepsilon/\theta > -1$, and by item c)

$$\limsup_{n\to\infty} J_n(-\varepsilon, P_n) \le e^{-1} < e^{-\varepsilon/\theta} = J(-\varepsilon, P).$$

$J(\cdot, P)$ is continuous, so convergence in distribution would require $J_n(x, P_n) \to J(x,P)$ at every x. It fails at $x = -\varepsilon$, so $J_n(\cdot, P_n)$ does not converge to $J(\cdot, P)$.

e) The bootstrap estimates $J_n(\cdot, P)$ by $J_n(\cdot, \hat{P}_n)$, where $\hat{P}_n$ is the empirical distribution. Since P is continuous, the n observations are distinct with probability one, so $\hat{P}_n$ puts mass $1/n$ on n distinct points of $[0,\theta]$, and $\theta(\hat{P}_n) = X_{(n)}$. That is, $\hat{P}_n$ is exactly a distribution of the kind studied in items c) and d), and the argument there applies conditional on the data. The bootstrap root is $R_n^* = n\big(X_{(n)}^* - X_{(n)}\big)$, and items c)–d) show that its distribution does not approach $J(\cdot, P)$. In fact, the bootstrap root equals zero whenever some bootstrap draw hits the sample maximum, so

$$\mathbb{P}^*\big(R_n^* = 0\big) = 1 - \left(1 - \frac{1}{n}\right)^n \longrightarrow 1 - e^{-1} \approx 0.63,$$

while the true limit $-\theta X$ has no mass at zero. The bootstrap distribution keeps a large atom at zero; the true distribution never does.

The reason is that bootstrap consistency needs $J_n(\cdot, P_n) \to J(\cdot, P)$ along sequences $P_n \to P$ that include the empirical distribution. The maximum depends on the extreme right tail, exactly where $\hat{P}_n$ is a poor approximation of P: $\hat{P}_n$ has no mass above $X_{(n)}$ and a mass of $1/n$ at $X_{(n)}$, while P has a density there. Confidence intervals for θ built from this bootstrap do not have the right coverage, even asymptotically.

Problem 5

a) Given x_i, the term $\alpha + \beta x_i$ is a constant and $\epsilon_i \mid x_i \sim N(0, 2 + x_i^2)$, so

$$y_i \mid x_i \sim N\big(2 + 5x_i,\ 2 + x_i^2\big).$$

The conditional mean is linear, so OLS is consistent for (α, β); the conditional variance grows with x_i^2, so the errors are heteroskedastic. We read $x \sim N(0, 3)$ as variance 3.

b, c) Figure 2 (left) shows one simulated sample of size $n = 100$. OLS gives $\hat{\alpha} = 2.169$ and $\hat{\beta} = 4.893$. The fanning out of the points as $|x|$ grows is the heteroskedasticity.

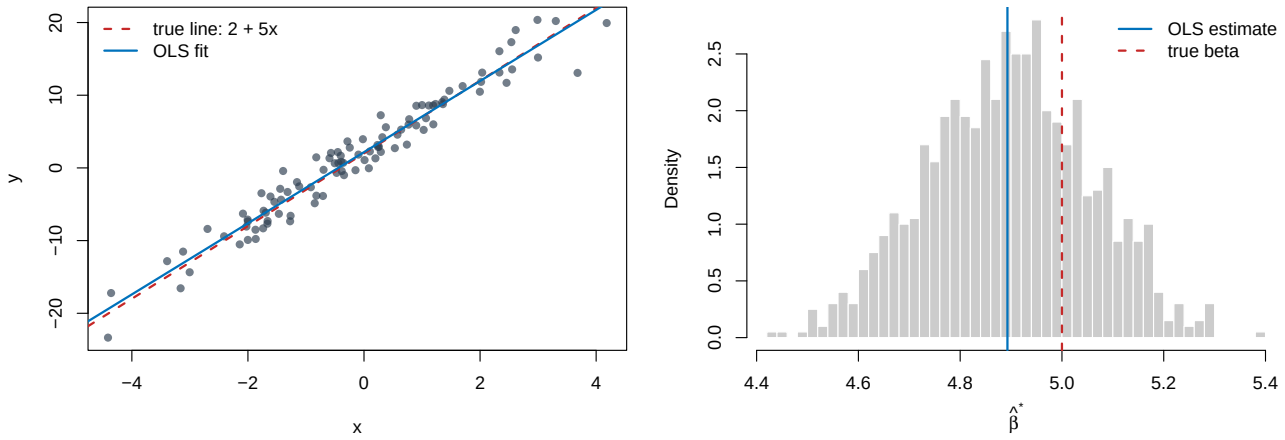


Figure 2: Left: one sample with the true line and the OLS fit. Right: the $B = 1000$ bootstrap estimates $\hat{\beta}^*$.

d) The three intervals are $\hat{\beta} \pm 1.96 \cdot \text{se}$:

- the usual OLS standard error, $\hat{s}^2 (X'X)^{-1}_{22}$ with $\hat{s}^2 = \sum \hat{\epsilon}_i^2 / (n-2)$;
- the heteroskedasticity-robust (HC1) standard error, from $\frac{n}{n-2} (X'X)^{-1} \big(\sum_i \hat{\epsilon}_i^2 x_i x_i' \big) (X'X)^{-1}$;

- the bootstrap standard error: the standard deviation of $\hat{\beta}^*$ over $B = 1000$ pairs-bootstrap samples, each drawing n pairs (y_i, x_i) with replacement. Resampling pairs keeps each error with its own x_i, so it keeps the heteroskedasticity.

	Standard error	95% confidence interval	
OLS (homoskedastic)	0.115	4.668	5.118
Robust (HC1)	0.158	4.584	5.202
Bootstrap (pairs, $B = 1000$)	0.159	4.581	5.205

The robust and bootstrap standard errors are almost identical, and about 40% larger than the usual one. The histogram of $\hat{\beta}^*$ in Figure 2 (right) is bell shaped and centered at $\hat{\beta}$, not at $\beta = 5$: the bootstrap reproduces the spread of $\hat{\beta}$, not its location.

e) Over $m = 500$ simulated samples:

	Coverage of $\beta = 5$	Average length
OLS (homoskedastic)	80.6%	0.503
Robust (HC1)	92.4%	0.714
Bootstrap (pairs)	93.0%	0.708

The usual interval is far too short. We can predict its coverage. The true asymptotic variance of $\hat{\beta}$ is the sandwich

$$\frac{1}{n}\frac{\mathbb{E}[x^2(2+x^2)]}{(\mathbb{E}[x^2])^2} = \frac{1}{n}\frac{2\cdot 3 + 27}{9} = \frac{3.67}{n},$$

using $\mathbb{E}[x^4] = 3 \cdot 3^2 = 27$, while the homoskedastic formula estimates $\frac{1}{n}\frac{\mathbb{E}[2+x^2]}{\mathbb{E}[x^2]} = \frac{1.67}{n}$. The usual standard error is too small by a factor $\sqrt{1.67/3.67} = 0.67$, so its nominal 95% interval actually covers with probability about $2\Phi(1.96 \cdot 0.67) - 1 \approx 0.81$, which is what we find. The variance of ϵ_i is largest exactly where x_i is far from its mean, the observations that drive $\hat{\beta}$, so ignoring heteroskedasticity understates the uncertainty.

The robust and bootstrap intervals both account for heteroskedasticity and cover close to 95%. The remaining shortfall is a small-sample effect: with $n = 100$, the sandwich is estimated with some downward bias, because the observations with large $|x_i|$ (high leverage) are also the ones with the largest error variance, and OLS residuals at high-leverage points are too small on average. The two methods target the same asymptotic variance, so they perform alike.

Listing 2: Problem 5 (`code/p5_ols_bootstrap.R`)

```
# Problem 5: OLS under heteroskedasticity, three confidence intervals for beta
```

```r
# Run from solutions/PS1/:  Rscript code/p5_ols_bootstrap.R

set.seed(2026)
alpha <- 2; beta <- 5; n <- 100; B <- 1000; m <- 500

# x ~ N(0, 3) with 3 the variance; eps | x ~ N(0, 2 + x^2)
simulate <- function(n) {
  x <- rnorm(n, 0, sqrt(3))
  y <- alpha + beta * x + rnorm(n, 0, sqrt(2 + x^2))
  data.frame(x = x, y = y)
}

ols <- function(x, y) {
  Xm <- cbind(1, x)
  XtXi <- solve(crossprod(Xm))
  b <- XtXi %*% crossprod(Xm, y)
  list(b = drop(b), e = drop(y - Xm %*% b), Xm = Xm, XtXi = XtXi)
}

# Receives a sample and returns the estimates and the three 95% intervals for beta
three_cis <- function(d, B = 1000) {
  fit <- ols(d$x, d$y)
  n <- nrow(d); k <- 2
  se_ols <- sqrt(sum(fit$e^2) / (n - k) * fit$XtXi[2, 2])
  meat <- crossprod(fit$Xm * fit$e)                # sum of e_i^2 x_i x_i'
  se_hc <- sqrt((fit$XtXi %*% meat %*% fit$XtXi)[2, 2] * n / (n - k))   # HC1
  b_star <- replicate(B, {                         # pairs (nonparametric) bootstrap
    i <- sample.int(n, n, replace = TRUE)
    ols(d$x[i], d$y[i])$b[2]
  })
  se_bs <- sd(b_star)
  z <- qnorm(0.975)
  se <- c(ols = se_ols, robust = se_hc, bootstrap = se_bs)
  list(b = fit$b, se = se,
       lo = fit$b[2] - z * se, hi = fit$b[2] + z * se, b_star = b_star)
}

# b)-d) one sample
d <- simulate(n)
res <- three_cis(d, B)

pdf("figures/p5_scatter.pdf", width = 6, height = 4.2)
par(mar = c(4.2, 4.2, 1, 1))
plot(d$x, d$y, pch = 16, col = "#2D3E50AA", xlab = "x", ylab = "y")
abline(alpha, beta, col = "#C62828", lwd = 2, lty = 2)
abline(res$b[1], res$b[2], col = "#0072BC", lwd = 2)
legend("topleft", bty = "n", lwd = 2, lty = c(2, 1), col = c("#C62828", "#0072BC"),
```

```
       legend = c("true␣line:␣2␣+␣5x", "OLS␣fit"))
dev.off()

pdf("figures/p5_bootstrap_hist.pdf", width = 6, height = 4.2)
par(mar = c(4.2, 4.2, 1, 1))
hist(res$b_star, breaks = 40, col = "grey80", border = "white", freq = FALSE,
     xlab = expression(hat(beta)^"*"), main = "")
abline(v = res$b[2], col = "#0072BC", lwd = 2)
abline(v = beta, col = "#C62828", lwd = 2, lty = 2)
legend("topright", bty = "n", lwd = 2, lty = c(1, 2), col = c("#0072BC", "#C62828"),
       legend = c("OLS␣estimate", "true␣beta"))
dev.off()

cat(sprintf("alpha_hat␣=␣%.3f,␣beta_hat␣=␣%.3f\n", res$b[1], res$b[2]))
print(round(cbind(se = res$se, lo = res$lo, hi = res$hi), 3))

# e) coverage over m = 500 samples
cover <- t(replicate(m, {
  r <- three_cis(simulate(n), B)
  c(cov = (r$lo <= beta & beta <= r$hi), len = r$hi - r$lo)
}))
cat("Coverage␣and␣average␣length␣over", m, "samples:\n")
print(round(colMeans(cover), 3))
```

Problem 6

a) The function `kde_ci` at the end of this problem does it. Two implementation details are worth noting. First, h is computed once from the original sample and reused in every bootstrap sample. Second, $\hat{f}_n(x)$ is the sample mean of $k_i = K\big((X_i - x)/h\big)/h$, and with h fixed, resampling the X_i's amounts to resampling the k_i's. So the code computes the k_i's once and each $\hat{f}^*_{n,b}(x)$ is the mean of n draws, with replacement, from them. This is the same bootstrap, only faster. The interval is

$$\Big[\hat{f}_n(x) - L_n^{-1}(0.975),\ \hat{f}_n(x) - L_n^{-1}(0.025)\Big].$$

b) With $n = 500$, $m = 1000$ samples and $B = 1000$:

	Coverage	Average width	Bias of $\hat{f}_n(x)$	s.d. of $\hat{f}_n(x)$	Bias/s.d.
$x = 0$	83.2%	0.081	-0.0168	0.0219	-0.77
$x = 1$	94.6%	0.071	-0.0005	0.0181	-0.03

Here $f(0) = 0.399$ and $f(1) = 0.242$. The bias and s.d. columns are computed across the m simulated

samples. The interval at $x = 1$ has the right coverage; the one at $x = 0$ misses $f(0)$ about one time in six.

c) For the standard normal, $f'(x) = -x\phi(x)$ and

$$f''(x) = (x^2 - 1)\phi(x), \qquad f''(0) = -\phi(0) = -0.399, \qquad f''(1) = 0.$$

By Problem 1 e), the bias is about $\frac{h^2}{2} f''(x)$. At $x = 0$, the mode, the density is concave and $\hat{f}_n(0)$ is biased downward; with $h \approx 1.06 \cdot 500^{-1/5} = 0.31$ the bias is about -0.019. (For the Gaussian kernel and normal data there is an exact formula: $\mathbb{E}[\hat{f}_n(x)]$ is the $N(0, 1 + h^2)$ density at x, which gives -0.017, as in the simulation.) This bias is about three quarters of the standard deviation of $\hat{f}_n(0)$. The interval is built around $\hat{f}_n(0)$, which is typically too low, so the whole interval is shifted down relative to $f(0)$ and covers less than 95%. At $x = 1$, an inflection point of ϕ, the leading bias vanishes, the bias is negligible relative to the standard deviation, and the coverage is close to the nominal level.

d) Conditional on the data, each bootstrap draw X_i^* is uniform over $\{X_1, \ldots, X_n\}$, so

$$\mathbb{E}^*\left[\frac{1}{h}K\left(\frac{X_i^* - x}{h}\right)\right] = \frac{1}{n}\sum_{j=1}^{n}\frac{1}{h}K\left(\frac{X_j - x}{h}\right) = \hat{f}_n(x), \qquad \text{hence} \qquad \mathbb{E}^*[\hat{f}_n^*(x)] = \frac{1}{n}\sum_{i=1}^{n}\hat{f}_n(x) = \hat{f}_n(x).$$

(The same h is used in both, which is why this holds exactly.) So the bootstrap roots $R_{n,b}^* = \hat{f}_{n,b}^*(x) - \hat{f}_n(x)$ have mean zero: the bootstrap "sees" no bias at all. In the bootstrap world the truth is $\hat{f}_n(x)$, and $\hat{f}_n^*(x)$ is unbiased for it. In the real world the truth is $f(x)$, and $\hat{f}_n(x)$ is unbiased only for the smoothed density $\mathbb{E}[\hat{f}_n(x)] = \int K(u) f(x + hu)\, du$. The bootstrap gets the variance right but not the bias, so the interval is a valid interval for the smoothed density, not for $f(x)$.

e) With $n = 2000$ at $x = 0$, the coverage is 86.2% (average width 0.050, bias -0.0103, s.d. 0.0129, so bias/s.d. $= -0.80$). More data does not fix it: the interval is narrower, but it is still centered in the wrong place. With Silverman's rule, $h \propto n^{-1/5}$, so

$$\text{bias} \approx \frac{h^2}{2}|f''(0)| \propto n^{-2/5}, \qquad \text{s.d.} \approx \sqrt{\frac{f(0)R(K)}{nh}} \propto n^{-2/5}.$$

Both shrink at the same rate, so their ratio does not shrink:

$$\frac{\text{bias}}{\text{s.d.}} \approx \frac{|f''(0)|}{2}\sqrt{\frac{nh^5}{f(0)R(K)}} = \frac{0.399}{2}\sqrt{\frac{1.06^5}{0.399 \cdot 0.282}} \approx 0.69,$$

with $R(\phi) = 1/(2\sqrt{\pi}) = 0.282$, whatever n is. A normal approximation then puts the coverage at $\Phi(1.96 - 0.69) - \Phi(-1.96 - 0.69) \approx 89\%$ even as $n \to \infty$. In our samples the ratio is a bit larger (0.77 and 0.80) because, with $h \approx 0.3$, the variance still has a non-negligible negative term $-\big(\mathbb{E}[\hat{f}_n(x)]\big)^2/n$ (see Problem 1 c)), which fades only slowly. So the coverage creeps up from 83% toward about 89%, and never reaches 95%. This is the role of the condition $nh^5 \to 0$ (rather than $nh^5 = O(1)$) from the lecture: Silverman's rule is built to minimize mean squared error, which balances squared bias and variance, so the bias stays of the same order as the standard deviation.

f) With $h/3$ (same rule, divided by 3, in the original and the bootstrap samples), at $n = 500$ and $x = 0$, the coverage is 93.6% and the average width is 0.169, against 83.2% and 0.081 with Silverman's h (the bias is -0.0002 and the s.d. 0.0431). To leading order, dividing h by 3 divides the bias by 9 and multiplies the standard deviation by $\sqrt{3} \approx 1.73$ (here by 2, as the negative $O(1/n)$ variance term matters less with a small h), so the bias-to-s.d. ratio falls by a factor of about 15, to about 0.05. The bias becomes negligible and the coverage is back near 95%. The price is precision: the interval is about twice as wide, and the estimator has a larger mean squared error (its s.d. alone, 0.043, exceeds the root-MSE with Silverman's h, $\sqrt{0.0168^2 + 0.0219^2} = 0.028$). This is the trade-off: a bandwidth that is good for estimation (small MSE) is too large for inference, and valid intervals require undersmoothing, at the cost of wider intervals.

Listing 3: Problem 6 (`code/p6_kde_bootstrap.R`)

```r
# Problem 6: bootstrap confidence intervals for a kernel density estimate
# Run from solutions/PS1/:  Rscript code/p6_kde_bootstrap.R

set.seed(2026)
B <- 1000; m <- 1000

# a) Bootstrap CI for f(x) from the root R_n = f_hat(x) - f(x).
# h is computed once, on the original sample, and reused in every bootstrap sample.
# hfac scales Silverman's h (hfac = 1/3 for item f).
kde_ci <- function(X, x, B = 1000, hfac = 1, level = 0.95) {
  n <- length(X)
  h <- hfac * 1.06 * sd(X) * n^(-1/5)
  k <- dnorm((X - x) / h) / h          # f_hat(x) is the mean of these n numbers
  f_hat <- mean(k)
  # Resampling X_i with replacement resamples k_i, so f*_hat(x) is the mean of the
  # resampled k's. Column b of idx holds the n draws of bootstrap sample b.
  idx <- matrix(sample.int(n, n * B, replace = TRUE), n, B)
  roots <- colMeans(matrix(k[idx], n, B)) - f_hat
  q <- quantile(roots, c((1 + level) / 2, (1 - level) / 2), names = FALSE)
  c(lo = f_hat - q[1], hi = f_hat - q[2], f_hat = f_hat, h = h)
}

coverage <- function(n, x, hfac = 1) {
  out <- t(replicate(m, kde_ci(rnorm(n), x, B, hfac)))
  f0 <- dnorm(x)
  c(n = n, x = x, hfac = hfac,
    coverage = mean(out[, "lo"] <= f0 & f0 <= out[, "hi"]),
    avg_width = mean(out[, "hi"] - out[, "lo"]),
    bias = mean(out[, "f_hat"]) - f0,
    sd = sd(out[, "f_hat"]),
    avg_h = mean(out[, "h"]))
}

res <- rbind(
  coverage(500, 0),            # b)
```

```
  coverage(500, 1),            # b)
  coverage(2000, 0),           # e)
  coverage(500, 0, hfac = 1/3) # f)
)
print(round(res, 4))

# Theory check, treating h as fixed at its population value (sigma = 1).
# With K = phi and X ~ N(0,1), E f_hat(x) is the N(0, 1 + h^2) density at x.
# sd_lead is the leading term sqrt(f R(K) / (nh)); sd_exact also keeps the -E[f_hat]^2/n term.
th <- function(n, x, hfac = 1) {
  h <- hfac * 1.06 * n^(-1/5)
  mean_f <- dnorm(x, 0, sqrt(1 + h^2))
  m2 <- integrate(function(u) dnorm(u)^2 * dnorm(x + h * u), -Inf, Inf)$value / h
  sd_exact <- sqrt((m2 - mean_f^2) / n)
  sd_lead <- sqrt(dnorm(x) / (2 * sqrt(pi)) / (n * h))
  bias <- mean_f - dnorm(x)
  c(h = h, bias = bias, bias_lead = h^2 / 2 * (x^2 - 1) * dnorm(x),
    sd_exact = sd_exact, sd_lead = sd_lead, ratio = bias / sd_exact,
    cov_normal = pnorm(1.96 - bias / sd_exact) - pnorm(-1.96 - bias / sd_exact))
}
print(round(rbind(th(500, 0), th(500, 1), th(2000, 0), th(500, 0, 1/3), th(1e8, 0)), 4))
```

Appendix: Consistency of the Local-Linear Estimator

Lecture 1 states the asymptotic distribution of the local-linear estimator and refers to Hansen's *Econometrics* for the proof, but it does not prove consistency. This appendix does, in two ways. The first is direct: write the estimator as a ratio of kernel averages and show that each average converges, as in Problem 1. The second treats the estimator as an extremum estimator. It uses the same averages, and it shows why the intercept recovers $m(x)$.

Setup. We observe (Y_i, X_i), $i = 1, \dots, n$. Let $m(z) = \mathbb{E}[Y \mid X = z]$ and $\mu_2(z) = \mathbb{E}[Y^2 \mid X = z]$. Fix a point x and write

$$u_i = \frac{X_i - x}{h}, \qquad K_i = K(u_i), \qquad \kappa_j = \int u^j K(u)\, du,$$

so $\kappa_0 = 1$, $\kappa_1 = 0$ by symmetry, and $\kappa_2 > 0$ is the kernel's variance. We assume:

(A1) (Y_i, X_i) are i.i.d. with $\mathbb{E}[Y^2] < \infty$. X has a density f with $M = \sup_z f(z) < \infty$; f is continuous at x and $f(x) > 0$.

(A2) m is continuous at x, and $\mu_2(z) f(z) \le C$ for all z.

(A3) K is a symmetric density, bounded by $\bar{K}$, with $\int u^4 K(u)\, du < \infty$.

(A4) $h \to 0$ and $nh \to \infty$.

These are weaker than what the asymptotic distribution needs: no derivatives of m or f, and no condition such as $nh^5 = O(1)$. The Gaussian and Epanechnikov kernels satisfy (A3). Condition (A2) holds, for example, if m and $\sigma^2(z) = \mathrm{Var}(Y \mid X = z)$ are bounded, since then $\mu_2 f \le (\sup m^2 + \sup \sigma^2) M$.

Proposition A.1. *Under (A1)–(A4),* $m_{LL}(x) \xrightarrow{p} m(x)$.

Proof 1: direct

Step 1: the estimator as a ratio of kernel averages. Define, for $j = 0, 1, 2$,

$$s_j = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h} K_i\, u_i^j, \qquad t_j = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h} K_i\, u_i^j Y_i.$$

Note that $s_0 = \hat{f}_n(x)$ is the kernel density estimator of Problem 1. In the closed form of Problem 2 a), the weights are $K_h(X_i - x) = K_i/h$ and $(X_i - x)^j = h^j u_i^j$, so $S_j = nh^j s_j$ and $T_j = nh^j t_j$. Numerator and denominator both pick up the factor $n^2 h^2$, which cancels:

$$m_{LL}(x) = \frac{S_2 T_0 - S_1 T_1}{S_0 S_2 - S_1^2} = \frac{s_2 t_0 - s_1 t_1}{s_0 s_2 - s_1^2}. \tag{2}$$

(When the denominator is zero the estimator is undefined; give it any value. We show below that this happens with probability going to zero.)

Step 2: each average converges.

Lemma A.2. *Under (A1)–(A4),* $s_j \xrightarrow{p} \kappa_j f(x)$ *for* $j = 0, 1, 2$, *and* $t_j \xrightarrow{p} \kappa_j\, m(x) f(x)$ *for* $j = 0, 1$.

Proof. As in Problem 1 d), we show that the mean converges and the variance goes to zero; Chebyshev's inequality then gives convergence in probability.

The s_j's. By the change of variables (1),

$$\mathbb{E}[s_j] = \mathbb{E}\left[\frac{1}{h} K_i u_i^j\right] = \int K(u) u^j f(x+hu)\, du, \qquad \text{so} \qquad \mathbb{E}[s_j] - \kappa_j f(x) = \int K(u) u^j \big[f(x+hu) - f(x)\big]\, du.$$

For each u the integrand goes to zero as $h \to 0$, because f is continuous at x. It is bounded by $2M|u|^j K(u) \le 2M(1+u^2)K(u)$, which is integrable. By dominated convergence, $\mathbb{E}[s_j] \to \kappa_j f(x)$. For the variance, using independence, $K^2 \le \bar{K} K$, and the same change of variables,

$$\operatorname{Var}(s_j) \le \frac{1}{n}\mathbb{E}\left[\frac{1}{h^2} K_i^2 u_i^{2j}\right] = \frac{1}{nh}\int K^2(u) u^{2j} f(x+hu)\, du \le \frac{M\bar{K}}{nh}\int (1+u^4) K(u)\, du \;\longrightarrow\; 0,$$

since $nh \to \infty$.

The t_j's. Let $g(z) = m(z) f(z)$. By the law of iterated expectations, $\mathbb{E}[K_i u_i^j Y_i \mid X_i] = K_i u_i^j m(X_i)$, so

$$\mathbb{E}[t_j] = \mathbb{E}\left[\frac{1}{h} K_i u_i^j m(X_i)\right] = \int K(u) u^j g(x+hu)\, du.$$

The function g is continuous at x, as a product of functions continuous at x. It is also bounded: by Jensen's inequality $m(z)^2 \le \mu_2(z)$, so $|g(z)| \le \sqrt{\mu_2(z) f(z)}\,\sqrt{f(z)} \le \sqrt{CM}$. The dominated convergence argument used for s_j then gives $\mathbb{E}[t_j] \to \kappa_j g(x)$. For the variance, iterated expectations again turn Y_i^2 into $\mu_2(X_i)$:

$$\begin{aligned}\operatorname{Var}(t_j) \le \frac{1}{n}\mathbb{E}\left[\frac{1}{h^2} K_i^2 u_i^{2j} \mu_2(X_i)\right] &= \frac{1}{nh}\int K^2(u) u^{2j} \mu_2(x+hu) f(x+hu)\, du \\ &\le \frac{C\bar{K}}{nh}\int (1+u^2) K(u)\, du \;\longrightarrow\; 0. \qquad \square\end{aligned}$$

Step 3: the ratio converges. By Lemma A.2 and the continuous mapping theorem,

$$s_0 s_2 - s_1^2 \xrightarrow{p} \kappa_2 f(x)^2, \qquad s_2 t_0 - s_1 t_1 \xrightarrow{p} \kappa_2 f(x) \cdot m(x) f(x) - 0 = \kappa_2\, m(x) f(x)^2.$$

The limit of the denominator is strictly positive, since $\kappa_2 > 0$ and $f(x) > 0$. So the probability that the denominator is positive (that is, that $Z'W^{-1}Z$ in Problem 2 a) is invertible) goes to one. The map $(a, b) \mapsto a/b$ is continuous wherever $b \neq 0$, so by the continuous mapping theorem and (2),

$$m_{LL}(x) \xrightarrow{p} \frac{\kappa_2\, m(x) f(x)^2}{\kappa_2 f(x)^2} = m(x). \qquad \square$$

What drives the result. The key step is $s_1 \xrightarrow{p} 0$. In the limit, the matrix $\begin{pmatrix} s_0 & s_1 \\ s_1 & s_2 \end{pmatrix}$ (which is $Z'W^{-1}Z$, rescaled) becomes $f(x)\,\mathrm{diag}(1, \kappa_2)$. It is diagonal because a symmetric kernel puts as much weight to the left of x as to the right, so the local regressor u_i is uncorrelated with the constant. Then the intercept no longer depends on the slope, and (2) reduces to t_0/s_0, the Nadaraya–Watson estimator, whose limit is $g(x)/f(x) = m(x)$. In finite samples $s_1 \neq 0$, and the slope correction is what removes the design bias discussed in Problem 2 d). It does not matter for consistency.

Proof 2: as an extremum estimator

Lecture 1 defines $m_{LL}(x)$ as the minimizer of a weighted least-squares criterion, so it is an extremum estimator. To use the consistency theorem for extremum estimators, we need a criterion with a non-degenerate limit, which takes two adjustments, since the criterion changes with n through h:

(i) *Divide by nh.* Only a share of about $2hf(x)$ of the observations gets non-negligible weight, so the average of the weighted squared residuals goes to zero. Dividing by nh instead of n keeps it of order one.

(ii) *Measure the slope in units of the bandwidth.* Over a window of width h, a line with slope b_1 rises by about hb_1. That is the quantity the local data can pin down, so we write $c = hb_1$ and $b_1(X_i - x) = c\,u_i$.

The criterion is then

$$Q_n(a, c) = \frac{1}{nh} \sum_{i=1}^{n} K_i \big(Y_i - a - c\,u_i\big)^2.$$

It is the lecture's criterion divided by nh, with the slope reparametrized. So its minimizer is $a = m_{LL}(x)$ and $c = h\,m'_{LL}(x)$, where $m'_{LL}(x)$ is the estimated local slope. Expanding the square,

$$Q_n(a, c) = A_n - 2a\,t_0 - 2c\,t_1 + a^2 s_0 + 2ac\,s_1 + c^2 s_2, \qquad A_n = \frac{1}{nh} \sum_{i=1}^{n} K_i Y_i^2.$$

The term A_n does not depend on (a, c), so we drop it: $\tilde{Q}_n = Q_n - A_n$ has the same minimizer. (This also saves us from assuming anything about the limit of A_n.) For each fixed (a, c), $\tilde{Q}_n(a, c)$ is a fixed linear combination of the s_j's and t_j's, so Lemma A.2 gives

$$\tilde{Q}_n(a, c) \xrightarrow{p} \tilde{Q}(a, c) = f(x)\Big[a^2 + \kappa_2 c^2 - 2a\,m(x)\Big] = f(x)\Big[\big(a - m(x)\big)^2 + \kappa_2 c^2 - m(x)^2\Big].$$

Since $f(x) > 0$ and $\kappa_2 > 0$, the limit $\tilde{Q}$ has a unique minimum at $(a, c) = (m(x), 0)$.

The usual consistency theorem for extremum estimators needs a compact parameter space and uniform convergence of the criterion. Neither holds here: the parameter space is $\mathbb{R}^2$, and we only have conver-

gence at each point. Convexity makes up for both. The following is the minimization version of Theorem 2.7 in Newey and McFadden (1994).[1]

Theorem (Newey and McFadden, 1994, Theorem 2.7). *Suppose that (i) $Q_0(\theta)$ is uniquely minimized at θ_0; (ii) θ_0 is in the interior of a convex set Θ and $\hat{Q}_n(\theta)$ is convex on Θ; and (iii) $\hat{Q}_n(\theta) \xrightarrow{p} Q_0(\theta)$ for every $\theta \in \Theta$. Then, with probability going to one, $\hat{Q}_n$ has a minimizer $\hat{\theta}$ on Θ, and $\hat{\theta} \xrightarrow{p} \theta_0$.*

Take $\Theta = \mathbb{R}^2$, $\hat{Q}_n = \tilde{Q}_n$, $Q_0 = \tilde{Q}$ and $\theta_0 = (m(x), 0)$. Condition (i) was shown above. For (ii), $\tilde{Q}_n$ is a sum of squares with nonnegative weights, minus a constant, so it is convex. Condition (iii) is the pointwise limit above. Hence

$$m_{LL}(x) \xrightarrow{p} m(x) \qquad \text{and} \qquad h\, m'_{LL}(x) \xrightarrow{p} 0.$$

Two remarks. First, the two proofs do the same computation in a different order. Proof 1 minimizes first (the closed form) and then takes the limit; Proof 2 takes the limit of the criterion first and then minimizes. They agree because the criterion is quadratic. Second, $h\, m'_{LL}(x) \xrightarrow{p} 0$ says nothing about the slope estimate itself. In the limit criterion the slope enters only through $\kappa_2 c^2$, which is minimized at $c = 0$ whatever $m'(x)$ is: at the scale of the bandwidth, every smooth function looks flat. Estimating $m'(x)$ consistently needs a finer argument and a stronger condition on h (the variance of $m'_{LL}(x)$ is of order $1/(nh^3)$, not $1/(nh)$), which we do not pursue here.

[1] W. K. Newey and D. McFadden (1994), "Large Sample Estimation and Hypothesis Testing," in R. F. Engle and D. L. McFadden (eds.), *Handbook of Econometrics*, Vol. 4, Ch. 36.